

HDOC Day-32 - C Programming Solution

Question: Write a C program to perform menu-driven operations using separate functions: (1) take two variable-length 2D arrays and prove matrix addition is commutative; (2) take two variable-length 2D arrays and prove matrix multiplication is not commutative; (3) Exit.

Algorithm / Logic

1. Display a menu repeatedly. For addition, read two matrices of the same order, calculate A+B and B+A, display both and compare them. If all corresponding elements are equal, addition is commutative.
2. For multiplication, use square matrices of the same order so that both AB and BA are defined. Calculate and display AB and BA. If at least one corresponding element differs, the chosen matrices demonstrate that matrix multiplication is not commutative. If AB happens to equal BA for the entered pair, the program explains that this particular pair commutes; another pair is needed to demonstrate the general non-commutative property.

C Program

```
#include <stdio.h>

void readMatrix(int r, int c, int a[r][c], char name)
{
    int i, j;
    printf("Enter elements of matrix %c:\n", name);
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
}

void printMatrix(int r, int c, int a[r][c])
{
    int i, j;
    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++)
            printf("%5d", a[i][j]);
        printf("\n");
    }
}

void addMatrix(int r, int c, int a[r][c],
               int b[r][c], int sum[r][c])
{
    int i, j;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            sum[i][j] = a[i][j] + b[i][j];
}

int equalMatrix(int r, int c, int a[r][c], int b[r][c])
{
    int i, j;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            if (a[i][j] != b[i][j])
                return 0;
    return 1;
}

void multiplyMatrix(int n, int a[n][n],
                   int b[n][n], int p[n][n])
{
    int i, j, k;
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++) {
            p[i][j] = 0;
            for (k = 0; k < n; k++)
                p[i][j] += a[i][k] * b[k][j];
        }
}
```

```

void additionDemo(void)
{
    int r, c;
    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &c);

    int A[r][c], B[r][c], AB[r][c], BA[r][c];

    readMatrix(r, c, A, 'A');
    readMatrix(r, c, B, 'B');

    addMatrix(r, c, A, B, AB);
    addMatrix(r, c, B, A, BA);

    printf("\nA + B:\n");
    printMatrix(r, c, AB);
    printf("\nB + A:\n");
    printMatrix(r, c, BA);

    if (equalMatrix(r, c, AB, BA))
        printf("\nA+B = B+A. Matrix addition is commutative.\n");
}

void multiplicationDemo(void)
{
    int n;
    printf("Enter order of square matrices: ");
    scanf("%d", &n);

    int A[n][n], B[n][n], AB[n][n], BA[n][n];

    readMatrix(n, n, A, 'A');
    readMatrix(n, n, B, 'B');

    multiplyMatrix(n, A, B, AB);
    multiplyMatrix(n, B, A, BA);

    printf("\nA x B:\n");
    printMatrix(n, n, AB);
    printf("\nB x A:\n");
    printMatrix(n, n, BA);

    if (!equalMatrix(n, n, AB, BA))
        printf("\nA x B != B x A. Multiplication is not commutative.\n");
    else
        printf("\nFor this pair AB = BA. Try another pair; "
            "matrix multiplication is not commutative in general.\n");
}

int main(void)
{
    int choice;

    do {
        printf("\n--- MENU ---\n");
        printf("1. Prove matrix addition is commutative\n");
        printf("2. Demonstrate matrix multiplication is not commutative\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: additionDemo(); break;
            case 2: multiplicationDemo(); break;
            case 3: printf("Program ended.\n"); break;
            default: printf("Invalid choice.\n");
        }
    } while (choice != 3);

    return 0;
}

```

Example for Multiplication

Take $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. Then $AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$, while $BA = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix}$. Since AB is not equal to BA , this example demonstrates that matrix multiplication is not commutative in general.

Result

The program successfully verifies $A + B = B + A$ for matrix addition and can demonstrate $A \times B \neq B \times A$ for matrix multiplication using suitable input matrices.